

Patricia Medina

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Professional Experience

- Assistant Professor (tenure-track), Mathematics Department, New York City College of Technology (CUNY) 8/2022–present
- Assistant Professor (tenure-track), Computer Science Department, Yeshiva College, Yeshiva University 8/2019–5/2022
- Visiting Scholar, ICERM, Brown University Spring 2019 Semester
- Postdoctoral Scholar, Worcester Polytechnic Institute 2016–2019
- Research Assistant, NSF-DMS 1115827 “Hybrid modeling in porous media”, PI: Professor M. Peszyńska Summer 2015
- Instructor, Oregon State University 9/2014–5/2016
- Graduate Research Assistant, Oregon State University, supported by NSF-DMS 1115827 “Hybrid modeling in porous media”, PI: Professor M. Peszyńska 1/2012–6/2014
- Teaching Assistant, Oregon State University 2009–2013
- Teaching Assistant, Bowling Green State University 2006–2009
- Instructor at Universidad Simón Bolívar and Universidad Central de Venezuela 2003–2006

Education

- Ph.D., Mathematics, Oregon State University 5/2014
Dissertation: *Mathematical Treatment and Simulation of Methane Hydrates and Adsorption Models*.
Advisor: Dr. Malgorzata Peszyńska
- M.A., Mathematics, Bowling Green State University 5/2009
Successful completion of written comprehensive examination based on Mathematical Analysis and Abstract Algebra
- M.S., Mathematics, Universidad de los Andes 6/2003
Thesis: *Non-complementarity of Orlicz spaces in $L^1[0, 1]$ and $C[0, 1]$* .
Advisor: Dr. Diomedes Bárcenas
- B.Sc., Mathematics, Universidad Central de Venezuela 1/2001
Thesis: *Regression Analysis using local polynomials (non-parametric statistics)*.
Advisor: Dr. Ricardo Ríos

Research

Research Interests

My research lies at the intersection of applied mathematics, functional analysis, geometry, and machine learning. My current work focuses on developing mathematically grounded frameworks for machine learning, with particular emphasis on feature engineering and dimensionality reduction for LiDAR and 3D point-cloud data, measure-theoretic features such as Product Coefficients, and dynamical-system approaches to neural networks and autoencoders. In parallel, I am developing new research directions in functional analysis and geometric methods, including Orlicz spaces, curvature-based quantities, and Schwarzian-type constructions, with applications to data-driven models and reconstruction problems. More broadly, my research seeks to connect mathematical theory with computational methods for data-driven scientific problems.

Preprints and manuscripts under review

1. P. Medina and H. P. G. Lam (2026). “Depth-Dependent Hidden-State Collapse in Dynamical System Autoencoders for LiDAR Point-Cloud Classification.” *Submitted to the 2026 IEEE High Performance Extreme Computing Conference (HPEC 2026)*. arXiv:2607.14463 [cs.LG]. <https://arxiv.org/abs/2607.14463>

Refereed publications

1. P. Medina and R. Karkare (2026). “Integrating Product Coefficients for Improved 3D LiDAR Data Classification (Part II).” *Accepted for presentation at the 10th International Conference on Data Management, Analytics and Innovation (ICDMAI 2026)*, January 9–11, 2026, Kolkata, India; proceedings forthcoming.
2. P. Medina (2026). “Integrating Product Coefficients for Improved 3D LiDAR Data Classification.” In S. Goswami, S. Saha, K. Basu, & R. S. Beed (eds.), *Data Management, Analytics and Innovation*, Volume 2, pp. 369–372. *Lecture Notes in Networks and Systems*, vol. 1369. Springer. https://link.springer.com/chapter/10.1007/978-981-96-5860-2_21
3. F. P. Medina and R. Paffenroth, “Classification frameworks comparison on 3D point clouds,” *2021 IEEE High Performance Extreme Computing Conference (HPEC)*, 2021, pp. 1–6, doi: 10.1109/HPEC49654.2021.9622842.
4. F. P. Medina, R. Paffenroth (2021) “Machine Learning in LiDAR 3D Point Clouds.” In: Demir I., Lou Y., Wang X., Welker K. (eds) *Advances in Data Science*. Association for Women in Mathematics Series, vol 26. Springer, Cham. https://doi.org/10.1007/978-3-030-79891-8_6
5. F. Patricia Medina, “Machine Learning in Crowd Flow Exit Data,” Accepted, *2019 AWM Research Symposium Proceedings*, Association for Women in Mathematics Series.
6. Karamatou Yacoubou Djima, F. Patricia Medina, Linda Ness and Melanie Weber, “Heuristic Framework for Multi-Scale Testing of the Multi-Manifold Hypothesis,” pp. 47–80. In: Ellen Gasparovic and Carlotta Domeniconi (Eds.): *Research in Data Science*, Association of Women in Mathematics Series, Volume 17, Springer, 2019.
7. Anna Grim, Boris Iskra, Nianqiao Ju, Alona Kryshchenko, F. Patricia Medina, Linda Ness, Melissa Ngamini, Megan Owen, Randy Paffenroth, Sui Tang. “Analysis of Simulated Crowd

Flow Exit Data: Visualization, Panic Detection, Exit Time Convergence, Attribution, and Estimation,” pp. 239–281. In: Ellen Gasparovic and Carlotta Domeniconi (Eds.): *Research in Data Science*, Association of Women in Mathematics Series, Volume 17, Springer, 2019.

8. F. P. Medina and M. Peszyńska, “Hybrid Modeling and Analysis of Multicomponent Adsorption with Applications to Coalbed Methane,” in *Porous Media: Theory, Properties, and Applications*, Nova Science Publishers, Editor: Doris Wolfe. Chapter 1, pp. 1–52, 2016. ISBN 978-1-63485-474-0
9. M. Peszyńska, F. P. Medina, W. Hong, M. Torres, “Reduced Numerical Model for Methane Hydrate Formation under Conditions of Variable Salinity. Time-Stepping Variants and Sensitivity,” *Computation* 2016, Special Issue on Advances in Flow and Transport in Porous Media, 4(1), 1. <https://doi.org/10.3390/computation4010001>
10. N. Gibson, P. Medina, M. Peszyńska, R. Showalter, “Evolution of phase transitions in methane hydrate,” *J. Math. Anal. Appl.*, Volume 409, Issue 2 (2014), pp. 816–833, doi: 10.1016/j.jmaa.2013.07.023.
11. F. P. Medina and M. Peszyńska, “Stability for implicit-explicit schemes for non-equilibrium kinetic systems in weighted spaces with symmetrization,” *Journal of Computational and Applied Mathematics*, Volume 328 (2018), pp. 216–231. <https://doi.org/10.1016/j.cam.2017.07.020>

Theses

1. F. Patricia Medina, *Mathematical Treatment and Simulation of Methane Hydrates and Adsorption Models*, Ph.D. Dissertation, Oregon State University, 2014, Corvallis, Oregon, USA.
2. F. Patricia Medina, *Non-complementarity of Orlicz spaces in $L^1[0, 1]$ and $C[0, 1]$* , Master’s Thesis, Universidad de Los Andes, 2003, Mérida, Venezuela.
3. F. Patricia Medina, *Regression Analysis using local polynomials*, Licenciado Thesis, Universidad Central de Venezuela, 2000, Caracas, Venezuela.

Invited Talks and Conference Presentations

1. (Contributed Talk) *Geometric Data Representation Through Machine Learning and Dynamical Systems Viewpoints*. SIAM New England Section Annual Meeting, July 22, 2026.
2. (Invited Minisymposium Talk) *Building Community, Belonging, and Persistence Through a Math Circle at an Urban HSI*. SIAM Annual Meeting 2026, Cleveland, Ohio, July 6–10, 2026.
3. (Invited Colloquium) *Making Sense of 3D Point Clouds: Machine Learning and Mathematics*. Department of Mathematics, Marist University, April 3rd, 2026
4. (Contributed) *Enhancing 3D Point Cloud Classification with Measure-Theoretic Features: Integrating Product Coefficients into LiDAR Analysis (Part II)*. Ninth International Conference on Data Management, Analytics and Innovation (ICDMAI 2026), Kolkata, India January 10, 2026
5. *From Measure Theory to Machine Learning: Product Coefficients for LiDAR Data*. Speaker Series, CUNY Graduate Center October 17, 2025
6. (Invited Colloquium) *Enhancing Machine Learning with Mathematics: Better Mapping of the World*. Department of Mathematics, Kasetsart University, Bangkok, Thailand January 22, 2025

7. (Invited Lecture) *Engineering Features for Machine Learning on Multi-Class 3D Point Clouds*. Ninth International Conference on Data Management, Analytics and Innovation (ICDMAI 2025), Kolkata, India January 18, 2025
8. (Contributed) *Enhancing 3D Point Cloud Classification with Measure-Theoretic Features: Integrating Product Coefficients into LiDAR Analysis*. Ninth International Conference on Data Management, Analytics and Innovation (ICDMAI 2025), Kolkata, India January 18, 2025
9. (Invited) *Product Coefficients for Feature Engineering for ML Classification in Remote Sensing Data*. Seminario Internacional de Matemáticas November 29, 2024
10. (Invited Lecture) *Measure Theory and Machine Learning in Remote Sensing Data*. Mathematical Sciences Lecture Series, Rhode Island College September 19, 2024
11. (Invited Seminar Talk) *Measure Theory Quantities Applied to Machine Learning in LiDAR*. Department of Mathematics Seminar, Colgate University March 7, 2024
12. (Invited) *Product Coefficients: Mathematical Quantities Improving Machine Learning Frameworks*. Northwest Undergraduate Mathematics Symposium November 11, 2023
13. (Invited) *Product Coefficients: Mathematical Quantities Improving Machine Learning Frameworks*. Math Circles, Brigham Young University November 8, 2023
14. (Invited Seminar Talk) *Product Coefficients for Machine Learning on Multi-Class 3D Point Clouds*. Analysis Seminar, University of Rochester October 13, 2023
15. (Invited Tutorial) *Using Machine Learning Techniques to Explore and Analyze LiDAR 3D Point Clouds*. AMIGAs Tutorial 3, IPAM July 12, 2023
16. (Contributed) *Product Coefficients in Machine Learning in LiDAR*. The MAA Annual Meeting of the Metropolitan New York Section, Pace University April 29, 2023
17. (Workshop Talk) *City Tech–BYU Workshop (Part I)*. New York City College of Technology (CUNY), Brooklyn, NY April 4, 2023
18. (Invited) *How Math Can Help in Machine Learning for LiDAR 3D Point Clouds*. The MAA Annual Meeting of the Metropolitan New York Section May 1, 2022
19. (Contributed) *Machine Learning in LiDAR 3D Point Clouds*. AMS Special Session on *Mathematical Tools for Computer Vision Problems, II*, Joint Mathematics Meetings April 6, 2022 (co-author: Randy Paffenroth, Worcester Polytechnic Institute)
20. (Colloquium) *Classification Frameworks Comparison on 3D Point Clouds*. Department of Mathematics and Computer Science Colloquium, Queensborough Community College (CUNY), Bay-side, NY February 16, 2022 (co-author: Randy Paffenroth, Worcester Polytechnic Institute)
21. *Classification Frameworks Comparison on 3D Point Clouds* (F. Patricia Medina; co-author: Randy Paffenroth), IEEE HPEC 2021 2021
22. Two invited lectures, REU summer program “STEM for All”, University of Rochester, Rochester, NY 2021

23. (Invited) Deep learning in LiDAR and how mathematical ideas can help us in machine learning. Mathematics Graduate Student Seminar, California State University Channel Islands, Camarillo, CA 2019
24. (Postdoc/Grad seminar) Deep learning in LiDAR and how mathematical ideas can help us in machine learning. ICERM Graduate Student and Postdoc Seminar, ICERM, Brown University, Providence, RI 2019
25. (Invited Short Research Talk) Machine learning research in geospatial data. Algebraic Vision Research Cluster, ICERM, Brown University, Providence, RI 2019
26. (Invited) Deep learning in LiDAR and how mathematical ideas can help us in machine learning. Mathematics and Computer Science Colloquium, University of Dallas, Irving, TX 2019
27. (Invited) Deep learning in LiDAR and how mathematical ideas can help us in machine learning. Mathematics Department Colloquium, California Polytechnic State University, San Luis Obispo, CA 2019
28. (Research Group Presentation) Deep learning in LiDAR and how mathematical ideas can help us in machine learning. Prof. Randy Paffenroth Data Science Research group, WPI, Worcester, MA 2019
29. (Invited) Deep learning in LiDAR and how measure theory can help us in machine learning. Denksport series, WPI, Worcester, MA 2019
30. (Contributed) Deep Learning for classification of 3D point cloud LiDAR (joint with R. Paffenroth). SIAM Annual Meeting, Portland, OR 2018
31. (Contributed) Deep learning in 3D point cloud LiDAR data. AWIMS New England, Worcester, MA 2018
32. (Research Group Presentation) Analysis of Simulated Crowd Flow Exit Data: The Deep Learning Approach. Prof. Randy Paffenroth Data Science Research group, WPI 2018
33. (Research Group Presentation) Deep Learning in LiDAR Data. Prof. Randy Paffenroth Data Science Research group, WPI 2017
34. (Report progress) Project 2: Representation of Data as Multi-Scale Features and Measures. WiSDM Research Collaboration Workshop, ICERM, Providence, RI 2017
35. (Invited) Evolution of phase transitions in methane hydrate. Schlumberger Research Center, Cambridge, MA 2017
36. (Applied Math Days) Mathematical treatments and Simulation of a Methane Hydrate Model. Rensselaer Polytechnic Institute, Troy, NY 2017
37. (Contributed) Numerical Approximations for methane hydrate models. Women's Intellectual Research Symposium, Brown University, Providence, RI 2017
38. (Analysis and PDE Seminar) A Consequence of the absence of Dunford-Pettis Property in Orlicz Spaces. Department of Mathematical Sciences, WPI 2016
39. (Numerical Analysis Seminar) Numerical Approximation for a Model of Methane Hydrates. Department of Mathematical Sciences, WPI 2016
40. (Colloquium) Hybrid Modeling and Analysis of Multicomponent Adsorption with Applications to Coalbed Methane. Department of Mathematical Sciences, WPI 2016

41. (Finite Element Circus) Analysis and Numerical Approximations for Kinetic Adsorption Models. WPI 2016
42. (Poster Blitz) Systems of Conservation Laws for Thermodynamically consistent Adsorption. Workshop for Women in Math Sciences, SAMSI, Research Triangle Park, NC 2016
43. (Contributed) Analysis and Numerical Approximation for Adsorption Models. SIAM Geosciences, Stanford University, Stanford, CA 2015
44. (Applied Mathematics and Computation Seminar) New Stability Framework for Kinetic Systems. Oregon State University 2015
45. (Invited) Analysis and Numerical Approximation for Adsorption Models. SIAM CSE, Salt Lake City, UT 2015
46. (Applied Mathematics and Computation Seminar) Hyperbolic Systems for Adsorption. Cascade Computational and Applied Mathematics Seminar, Corvallis, OR 2014
47. (Invited) Numerical approximation for a model of Methane hydrates. Pacific Northwest Numerical Analysis Seminar, Seattle, WA 2013
48. (Contributed) Numerical approximation for a model of methane hydrates. Department of Mathematics, University of Padova; SIAM Geosciences, Padova, Italy 2013
49. (Applied Mathematics and Computation Seminar) Numerical Approximation for a model of methane hydrates. Oregon State University, Corvallis, OR 2013
50. (Applied Mathematics and Computation Seminar) Hyperbolic systems for adsorption (joint with M. Peszyńska). Oregon State University 2012
51. (Probability Seminar) The Dunford-Pettis theorem, Orlicz spaces and related topics. Oregon State University, Corvallis, OR 2010
52. (Probability Seminar) On the convergence in mean of martingale difference sequence. Oregon State University 2010
53. (Analysis Seminar) Uniform integrability and the De La Vallée Poussin and Dunford-Pettis theorems. Bowling Green State University, Bowling Green, OH 2008
54. (Contributed) XVII Jornadas Venezolanas de Matemáticas. Non-complementarity of Orlicz spaces in $L^1[0, 1]$ and $C[0, 1]$. Trujillo, Venezuela 2004
55. (Functional Analysis Seminar) Orlicz Spaces. Universidad de Los Andes, Mérida, Venezuela 2003
56. (Functional Analysis Seminar) Topology in $C[0, 1]$. Universidad de Los Andes, Mérida, Venezuela 2003

Posters

1. *The Neighbor Matrix: Feature Engineering and Dimension Reduction to Improve Machine Learning in 3D Point Clouds*. Patricia Medina, New York City College of Technology (CUNY). 22nd Annual Poster Session November 21, 2024
2. *The Neighbor Matrix: Feature Engineering Dimension Reduction to Improve ML in 3D Point Clouds* (with R. Paffenroth). 22nd Annual Poster Session November 16, 2023

3. *Product Coefficients on Dyadic Sets Applied to 3D Point Clouds* (with R. Karkare). 21st Annual Poster Session November 16, 2023
4. *Heuristic Framework for Multi-Scale Testing of the Multi-Manifold Hypothesis on 3D LiDAR Point Clouds* (with K. Y. Djima, L. Ness, and M. Weber). 20th Annual Poster Session. **Award:** Top 3 Best Poster. November 27, 2022
5. Mathematical treatment and simulation for a methane hydrate model. Modern Advances in Computational and Applied Mathematics (workshop honoring Charles L. Epstein and Leslie Greengard). Yale University, New Haven, CT 2017
6. Mathematical Treatment and simulation for methane hydrate models. High Performance Computing Days 2017. University of Massachusetts Dartmouth, North Dartmouth, MA 2017
7. Mathematical treatment and simulation for methane hydrate models. Frontiers in Applied and Computational Mathematics. Brown University, Providence, RI 2017
8. Hybrid Modeling and Analysis of Multicomponent Adsorption with Applications to Coalbed Methane. Blackwell-Tapia Conference and Award Ceremony. University of Tennessee, Knoxville, TN 2016
9. Systems of Conservation Laws for Thermodynamically consistent Adsorption. Workshop for Women in Math Sciences. SAMSI, Research Triangle Park, NC 2016
10. Systems of conservation laws for thermodynamically consistent adsorption with subscale diffusion and memory terms. Department of Mathematics, University of Padova; SIAM Geosciences, Padova, Italy 2013
11. Systems of conservation laws for thermodynamically consistent adsorption with sub-scale diffusion (joint with M. Peszyńska). SIAM Annual Meeting, Minneapolis, MN 2012

Grants and Awards

1. Grants

- PIs: F. Patricia Medina and Juan Li. Google Explore grant (\$15,000) 2020–2021
- P.I., AI for Earth Microsoft Azure Compute Grant (\$10,000). Project: “Solutions for climate change science: using deep learning to improve vegetation classification”. Co-PIs: Jonathan Batchelor, L. Monika Moskal, Randy Paffenroth, Guang Zhen 2018–2020
- P.A., Women Impact Network grant (WIN) with Lucia Carichino and Xiaodan Zhou to fund the event “Advancing Women’s Impact in Mathematics Symposium, New England (AWIMS)”, April 2018 (\$5,000) 2017
- Co-P.I., AWM for SK Days through NSF Grant NSF-DMS-1028861 to fund “Sonia Kovalevsky Days for Middle School Children” with Nathan Gibson, Holly Swisher, Vrushali Bokil and AWM student chapter officers, May 2011 (\$1,000) 2011

2. Participation in funded research

- Graduate Research Assistant, NSF-DMS 1115827 “Hybrid modeling in porous media”, PI: Professor M. Peszyńska 2012–2014
- Research Instructor, NSF-DMS 1115827 “Hybrid modeling in porous media”, PI: Professor M. Peszyńska Summer 2015

3. Travel grants, scholarships and distinctions

- **Established Mathematician Travel Grant**, American Mathematical Society (AMS), Expanding U.S. Mathematician Access Participation at the 2026 International Congress of Mathematicians (ICM 2026) Travel Grant Program, funded by National Science Foundation (NSF) Award #2620717, 2026. Award of up to \$2,400.
- AWM Travel grant for invited talk at AWM Research Symposium, Rice University, Houston 2018
- Microsoft Family Fellowship to support participation in the Computer Vision Long 2019 Spring Semester Program, ICERM, Brown University 2018
- ICERM travel support to participate in WiDS workshop research collaborations at ICERM 2017
- Partial support from AWM ADVANCE grant for travel expenses to participate in WiDS workshop at ICERM 2017
- NIMBioS Travel award. Blackwell-Tapia Conference and Award Ceremony. University of Tennessee, Knoxville, TN 2016
- SAMSI Travel award. Poster presentation. SAMSI, Research Triangle Park, NC 2016
- Early career SIAM Travel award. Contributed talk. Stanford University, Stanford, CA 2015
- Early career IMA Travel Award to attend New Directions Short Course on “Introduction to Uncertainty Quantification”. IMA, Minneapolis, MN 2015
- Early career SIAM Travel award to give an invited talk at SIAM CSE. Salt Lake City, UT 2015
- Student speaker at the Pacific Northwest Numerical Analysis Seminar (PNWNAS 2014), selected among students in all Pacific Northwest universities; nominated by Oregon State University 2014
- SIAM International student travel award to present at SIAM Geosciences. Padova, Italy 2013
- SIAM Student travel award for SIAM student chapter representative. SIAM Annual Meeting, Minneapolis, MN 2012
- OSU: nominated by the Department of Mathematics for attending MSRI summer workshop 2011
- MSRI Student travel award for MSRI summer graduate workshop. Berkeley, CA 2011
- Universidad Simón Bolívar partial funding for professors on leave for PhD studies 2006–2008
- Venezuelan Institute for Scientific Research: Travel Award to attend “Escuela Venezolana de Matemáticas” 2001
- FONACIT/CONICIT (Ministry of Science and Technology): Scholarship for graduate studies (Master in Science) 2001–2003
- Venezuelan Institute for Scientific Research: Scholarship for first two terms of graduate studies 2001

Professional Activities

- Co-organizer, lecturer and project leader, TRIPODS NSF REU–STEM FOR ALL 2021 Neural Networks, July 19–August 13 2021

- Lecturer and project leader, MAA–SIAM & TRIPODS Advanced Workshop in Data Science for Mathematical Sciences Faculty, Jun 28–Jul 2 2021
- Journal Referee, *Research in Data Science*, Association for Women in Mathematics Series, Springer 2021

Professional Development

- Geohack Week, eScience Institute, University of Washington, Seattle, WA 2018
- Women in Data Science and Mathematics Research Collaboration Workshop (WiDSM), ICERM, Providence, RI 2017
- Workshop for Women in Math Sciences, SAMSI, Research Triangle Park, NC 2016
- New Directions Short Course on “Introduction to Uncertainty Quantification”, IMA, Minneapolis, MN 2015
- Oregon State University Ecampus Faculty Forum, Corvallis, OR 2015
- IMA Special Workshop: Careers and Opportunities in Industry for Mathematical Scientists, Minneapolis, MN 2014
- Summer Graduate Workshop: “The Dirichlet Space: Connections between Operator Theory, Function Theory, and Complex Analysis”, MSRI, Berkeley, CA 2011
- Infinite Dimensional Analysis Conference, Kent State University, Kent, OH 2005
- Universidad Simón Bolívar graduate courses (Harmonic Analysis I and II), Caracas, Venezuela 2005–2006
- XVII Escuela Venezolana de Matemáticas y IV EMALCA, Mérida, Venezuela. Tensor products of Banach spaces (Joe Diestel and Johan Swart), Kent State University 2004
- XV Escuela Venezolana de Matemáticas y IV EMALCA, Mérida, Venezuela. I. Semigroups of operators and applications (Diomedes Bárcenas and Hugo Leiva, Universidad de los Andes). II. Ultraproducts in Analysis (Jose Iovino, UT San Antonio) 2002
- XIV Escuela Venezolana de Matemáticas, Mérida, Venezuela. Gaussian Processes extremes (Mario Wschebor, Universidad de la República, Uruguay) 2001

Advising and Mentoring

Independent Studies / Seminar Mentoring

- Evelyn Pulla, student researcher, *Data Exploration of Remote Sensing Indicators for Permafrost Thaw Risk*. Poster and lightning talk, 4th Annual Celebrating Research Excellence Symposium (CREST), New York City College of Technology, August 6, 2026.
- Faculty mentor, BYU–City Tech Data Science Seminar (independent studies; two semesters) Spring 2023 and Spring 2024

Student research supervision and outcomes

- *Sentiment Analysis of Instagram Comments: Exploring Body Image Perceptions* (with H. Bloomfield, Q. Chen, and M. Hughes). Poster, 22nd City Tech Annual Poster Session November 21, 2024

- *Comprehensive Crime Analysis: Integrating Classification, Regression, and Spatial-Temporal Insights* (with J. Mejia and B. Ndoni). Student project resulting in poster, talk, and written report 2024
- *NLP for Analyzing Mental Health Issues among College Students Using Reddit Data* (with Q. Chen, D. Dorsey, M. McGuire, and M. Hughes). Poster, 22nd Annual Poster Session November 16, 2023
- *Interdisciplinary Undergraduate Data Science Projects: A Joint Collaboration Between City Tech and Brigham Young University* (contributed poster). Contributed Poster Session, The MAA Annual Meeting of the Metropolitan New York Section, Pace University April 29, 2023 (with D. Gallego, K. Chang, Q. Chen, J. Corbridge, D. Dorsey, E. Lonsdale, A. McGuire, A. Oler, C. Razatos, and I. Roman Villanueva; advisors: M. Hughes, H. Carley, F. P. Medina, and C. Welker)

City Tech–BYU exchange workshops and student mentoring

- BYU–City Tech Workshop, City Tech (Brooklyn, NY) March 14–15, 2024
- BYU–City Tech Workshop, Brigham Young University (Provo, UT) May 23–25, 2024
- BYU–City Tech Workshop, City Tech (Brooklyn, NY) April 26–29, 2023
- BYU–City Tech Workshop, Brigham Young University (Provo, UT) May 30–June 1, 2023
- Faculty mentor for participating City Tech students: Q. Chen, J. Mejia, B. Ndoni, E. Peters, D. Gallego, C. Razatos 2023–2024

Service

- Invited and supported speaker event, City Tech SIAM Student Chapter: Dr. Ron Buckmire, *Different Differences: An Introduction to Nonstandard Finite Difference Schemes* March 12, 2026
- Invited and supported speaker event, City Tech SIAM Student Chapter: Dr. Megan Owen, *The Mathematics of Evolutionary Trees* February 19, 2026
- President and Co-Founder, CUNY SACNAS Professional Chapter, 2026–present.
- HSI Committee Member, New York City College of Technology (CUNY) Fall 2025–present
- Faculty advisor, City Tech SIAM Student Chapter Fall 2022–present
- Invited and supported speaker event, City Tech SIAM Student Chapter: Dr. Michael Dorff, *How Mathematics Is Making Hollywood Movies Better*. (Co-organized with U. Duttagupta and S. Singh.) April 4, 2024
- Co-organized City Tech Math Club talk (with J. Thiel): M. Hughes (BYU), *Can AI Learn to Perform Research in Pure Mathematics?* March 14, 2024
- Invited and supported speaker event, City Tech SIAM Student Chapter: Dr. Boris Iskra, *Navigating the Realm of Online Advertising Using Machine Learning* December 7, 2023
- Session chair, Contributed Talks, The MAA Annual Meeting of the Metropolitan New York Section, Pace University April 29, 2023

- Organizational Committee, “Advancing Women’s Impact in Mathematics Symposium, New England (AWIMS)”, Worcester, MA 2018
- Contributed talk session chair, Women’s Intellectual Research Symposium, Brown University, Providence, RI 2017
- Poster judge, Graduate Research Innovation Exchange (GRIE), Worcester, MA 2017
- Organizational committee, SIAM and AWM Student Chapter speaker series (Speaker: Dr. Talitha Washington, Howard University), Oregon State University 2014
- Cascade Computational and Applied Mathematics Seminar, student organizing committee, Corvallis, OR 2014
- OSU SIAM Student Chapter: President (2013–2014), Vice-President (2012–2013), Treasurer 2010–2011
- OSU AWM Student Chapter: President (2010–2011), Vice-President (2011–2012), Treasurer 2013
- Organizational committee, SIAM Student Chapter speaker series (Speaker: Dr. Thomas Grindine, Boeing; SIAM Vice-president for industry), Oregon State University 2013
- Member of panel, Q&A session for graduate students in Mathematics (MTH 599), Oregon State University 2013
- Chair, Sonia Kovalevskaya Days, Oregon State University, Corvallis, OR 2011
- Chair, student committee for faculty promotion and tenure, Department of Mathematics, Oregon State University 2010–2011

Technical Skills

- Programming: Python (NumPy, pandas, scikit-learn), MATLAB
- Machine Learning: PyTorch, TensorFlow
- Tools: Git/GitHub (basic)
- Scientific/Technical: $\text{\LaTeX}$
- Web: HTML/CSS
- Platforms: Linux/Unix, macOS, Windows

Languages

- Spanish (native)
- English (fluent)
- Italian (reading proficiency)
- French (reading proficiency)

Professional Associations

- AWM (Association for Women in Mathematics)
- AMS (American Mathematical Society)

- SIAM (Society of Industrial and Applied Mathematics)
- SACNAS (Society for Advancement of Chicanos/Hispanics and Native Americans in Science)